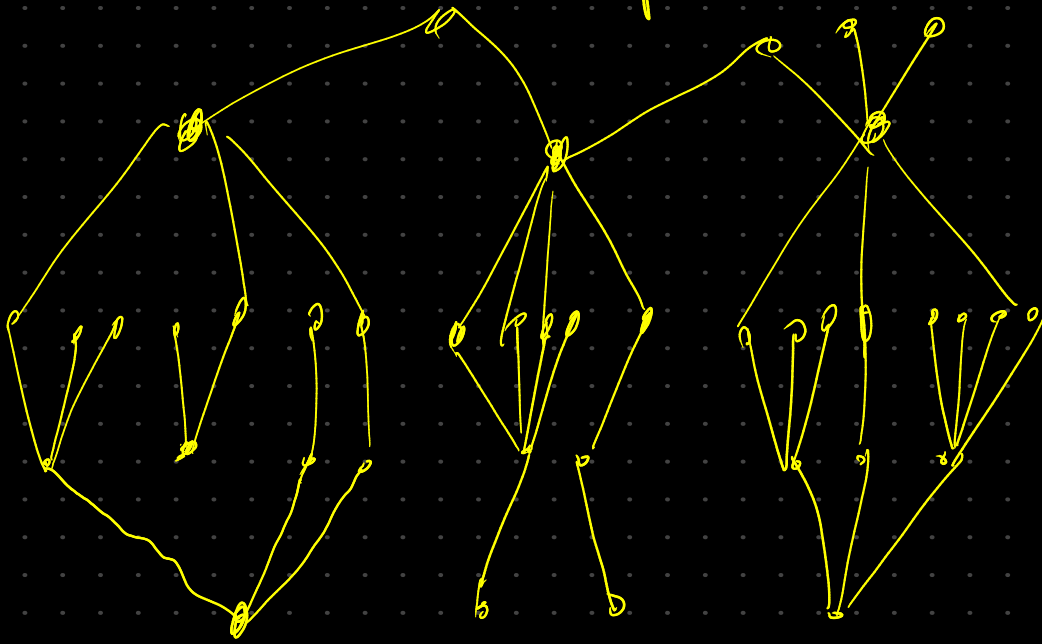
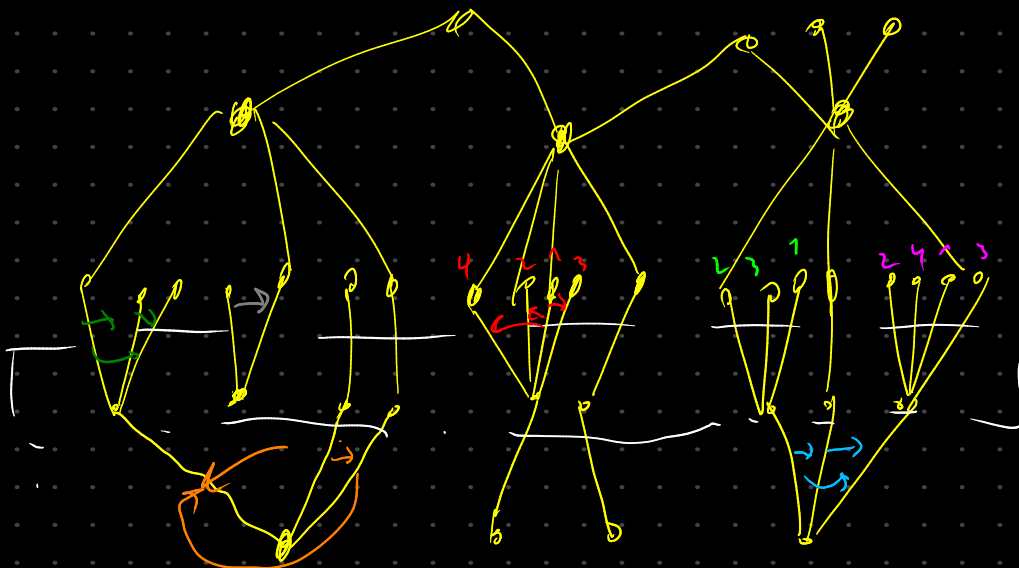
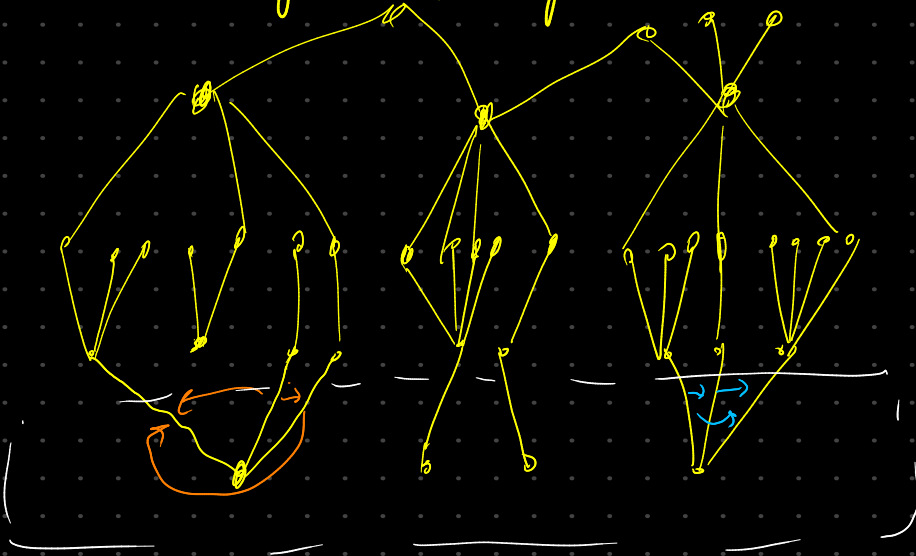


Goal: construction of the greedy strategy.



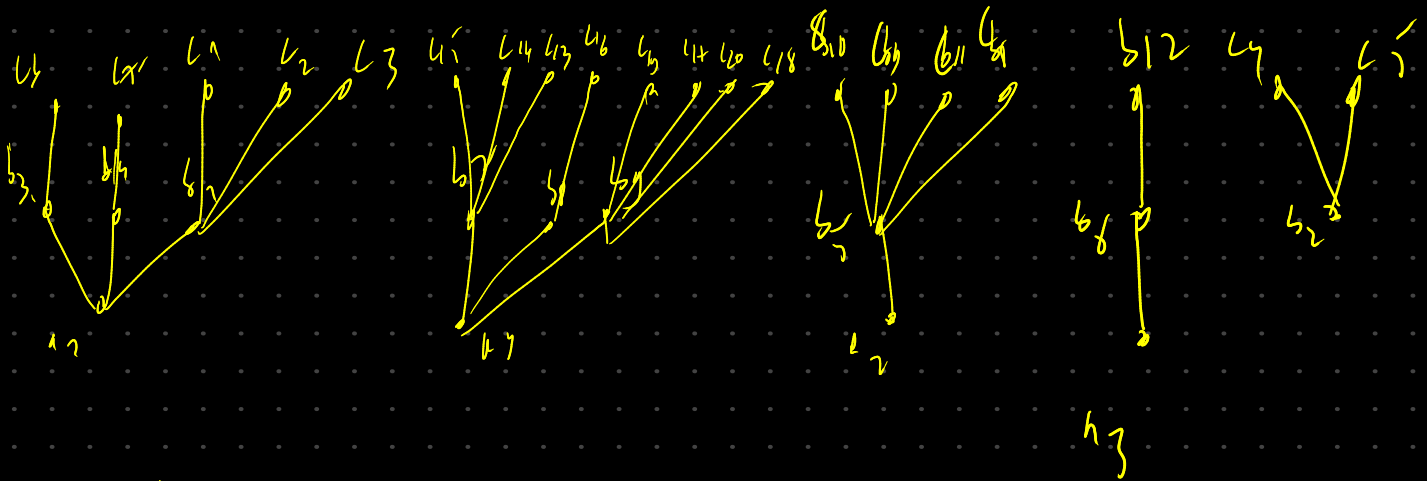
we have ϕ
first step order the adjacent edges of each level (non-cyclic way)



2) u_1, u_2, u_3

$b_4, b_5, b_6, b_7, b_8, b_9, b_{10}, b_{11}, b_{12}, b_{13}, b_{14}, b_{15}, b_{16}, b_{17}, b_{18}, b_{19}, b_{20}, b_{21}, b_{22}, b_{23}, b_{24}, b_{25}, b_{26}, b_{27}, b_{28}, b_{29}, b_{30}, b_{31}, b_{32}, b_{33}, b_{34}, b_{35}, b_{36}, b_{37}, b_{38}, b_{39}, b_{40}, b_{41}, b_{42}, b_{43}, b_{44}, b_{45}, b_{46}, b_{47}, b_{48}, b_{49}, b_{50}, b_{51}, b_{52}, b_{53}, b_{54}, b_{55}, b_{56}, b_{57}, b_{58}, b_{59}, b_{60}, b_{61}, b_{62}, b_{63}, b_{64}, b_{65}, b_{66}, b_{67}, b_{68}, b_{69}, b_{70}, b_{71}, b_{72}, b_{73}, b_{74}, b_{75}, b_{76}, b_{77}, b_{78}, b_{79}, b_{80}, b_{81}, b_{82}, b_{83}, b_{84}, b_{85}, b_{86}, b_{87}, b_{88}, b_{89}, b_{90}, b_{91}, b_{92}, b_{93}, b_{94}, b_{95}, b_{96}, b_{97}, b_{98}, b_{99}, b_{100}$

$b_{11}, b_{14}, b_{13}, b_{16}, b_{17}, b_{20}, b_{18}$



We found a vertex that gets more than an edge.

$d_1 \rightarrow$ incoming edges = $\{(c_1, d_1), (c_5, d_1), (c_7, d_1)\}$

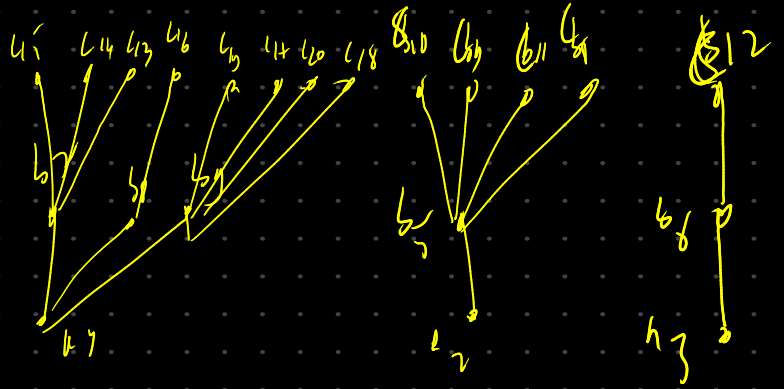
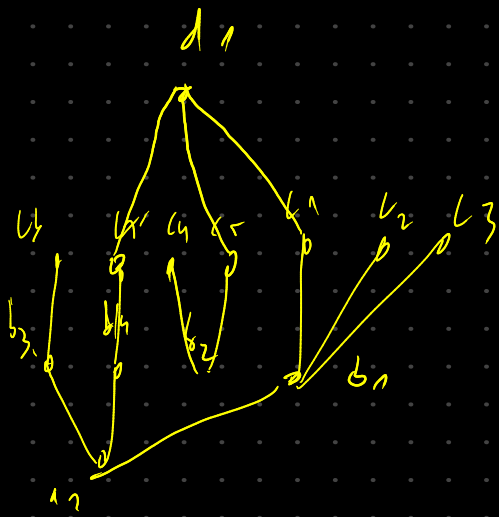
$\hookrightarrow$

c_1, c_5, c_7 no cycle!

already have c_1, c_7 $eq(c_1, c_7) = \{b_1, b_4\}$

$eq(c_1, c_5)$ no value

array $[c_1, c_7]$ out of adj.



we found $b_2 \rightarrow ((c_9, b_2), (c_{10}, b_2), (c_{11}, b_2), (c_{12}, b_2))$

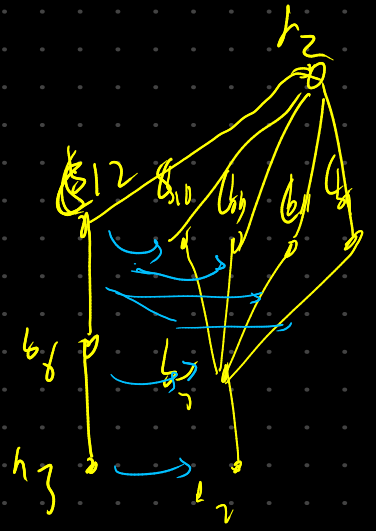
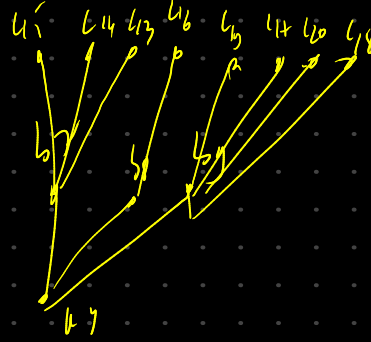
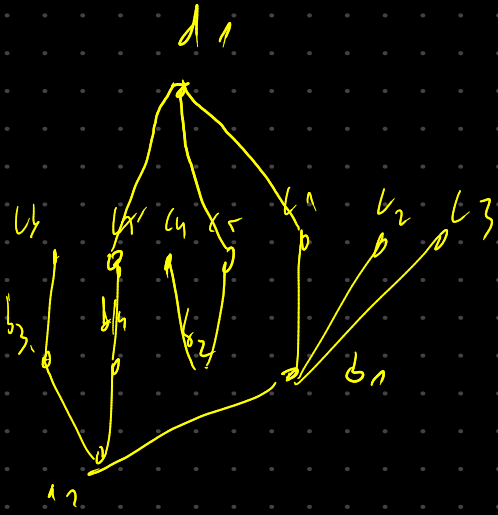
define non-cycle order of $c_9, c_{10}, c_{11}, c_{12}$

check eq (c_i, c_j) $i \neq j$ & $\{i, j\} \in \{9, 10, 11, 12\}$

$\{i, j\} \in \{9, 10, 11\} \Rightarrow \text{eq}(c_i, c_j)$ already set

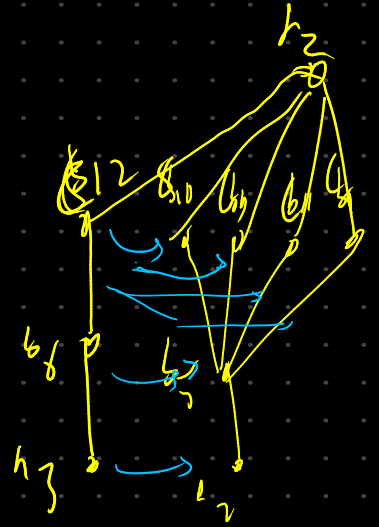
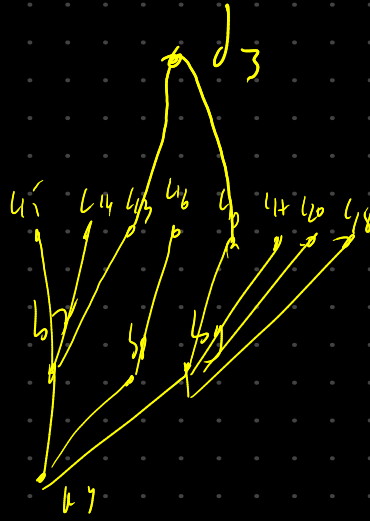
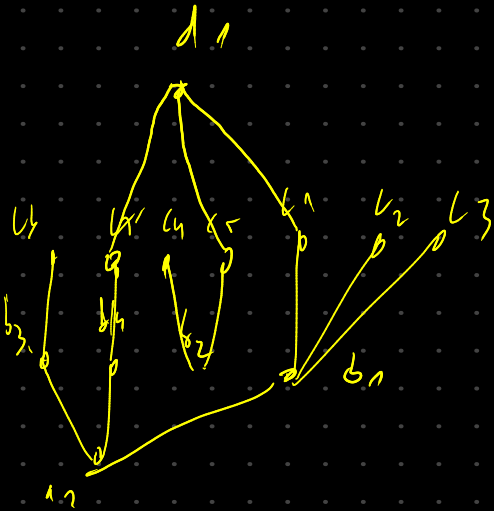
$[c_{10}, c_9, c_{11}, c_8]$ $[c_{12}]$

assign relation between the 2 lists if none is already assigned, in this case assigning $\phi(c_{10}, c_{12})$ will assign all $\phi(c_i, c_{12})$ $i \in \{c_9, c_8, c_{11}\}$



$d_3 (c_{13}, c_{15})$

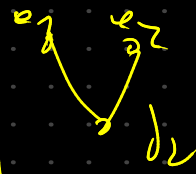
$e_1(c_{13}, c_{15}) = e_1(b_7, b_9)$ (empty set)

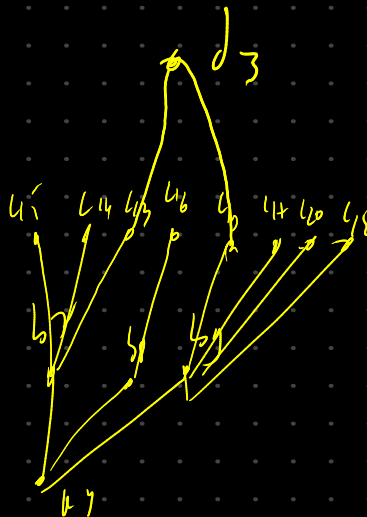
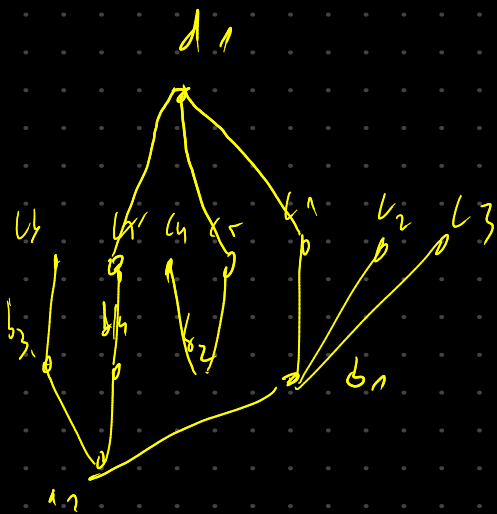


now we are looking for adjacent edges from the level (d) to (c)

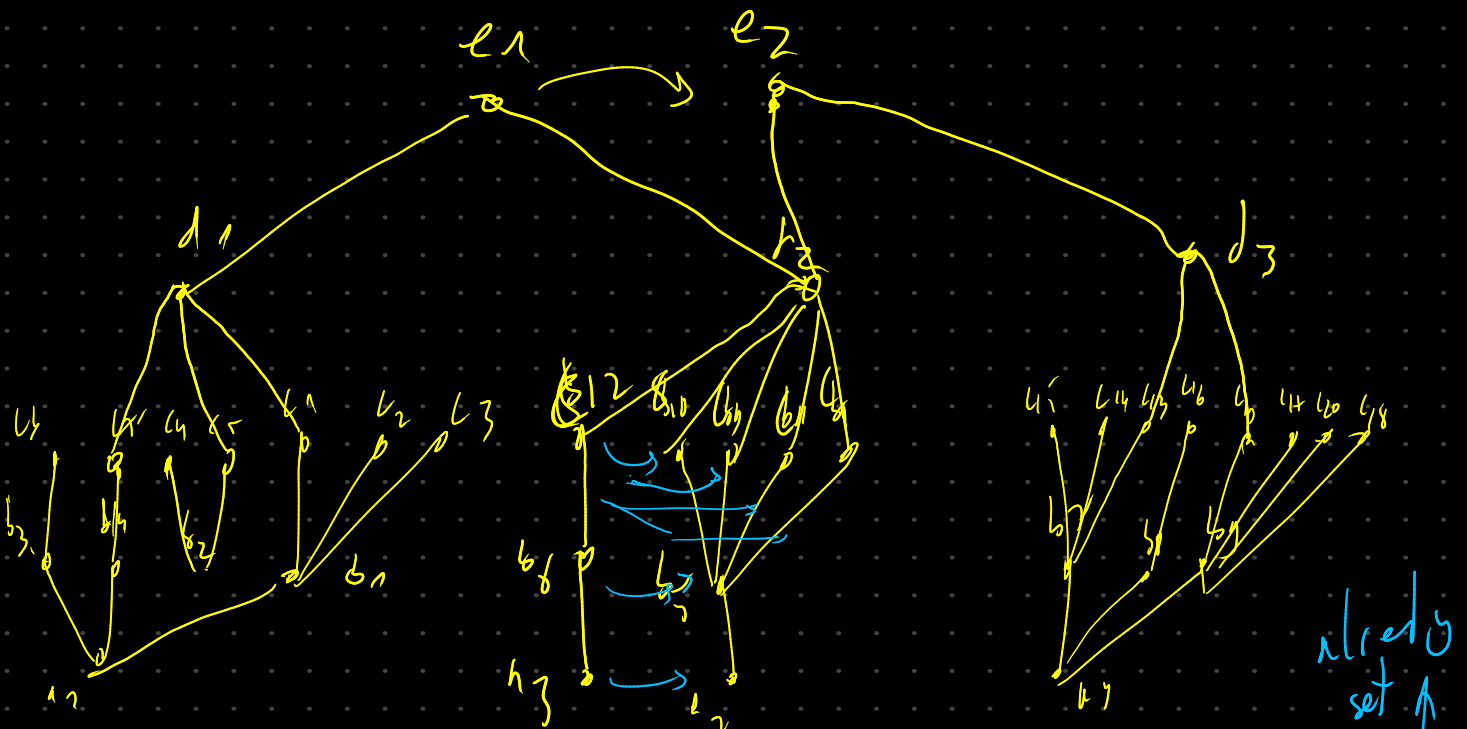
$e_1(e_1, e_2) = \emptyset \Rightarrow$

we have
pick arbitrary e_1, e_2





$$\phi(e_1, e_2) = \{ \phi(e_1, e_2), \dots, \phi(a_1, a_2), \phi(a_1, a_3), \phi(a_1, a_4), \phi(a_2, a_4), \phi(a_3, a_4) \}$$

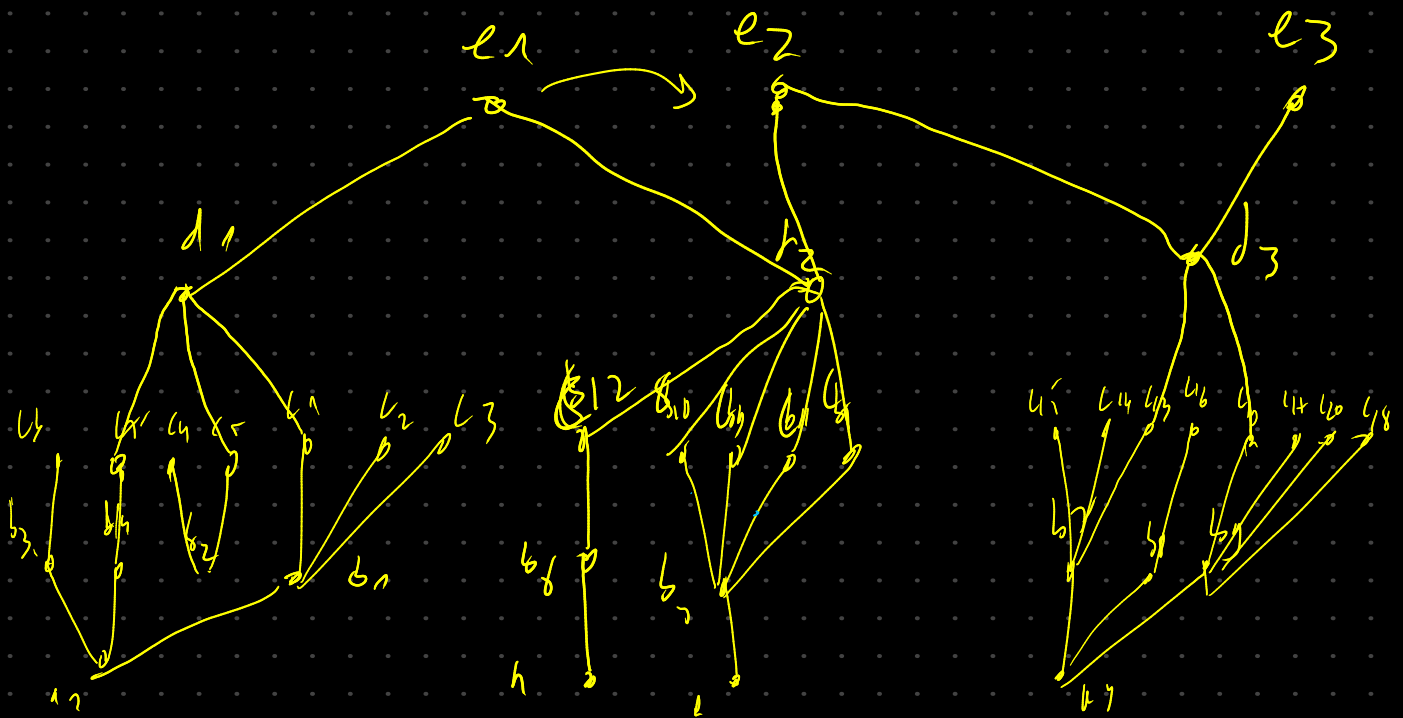


now d_3 e_2 e_3 e_4

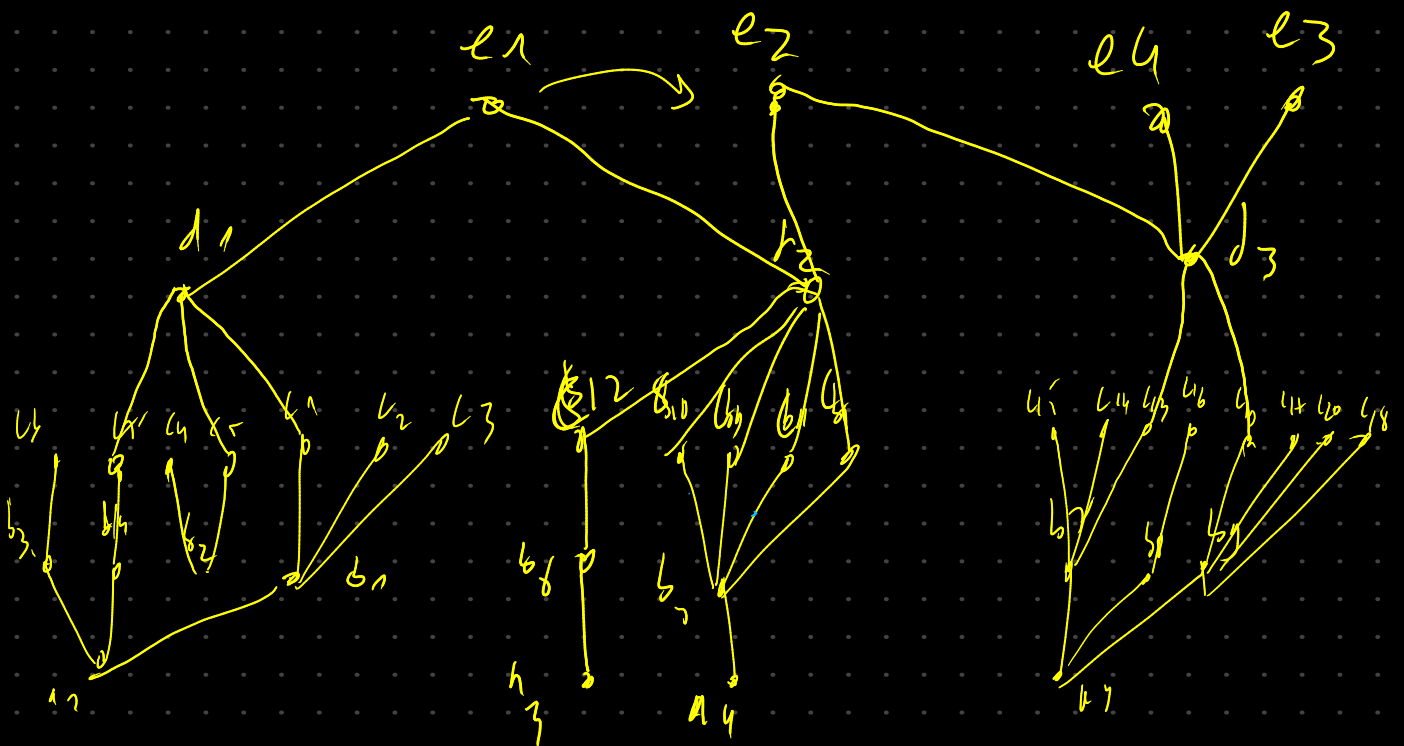
$$\phi(e_2, e_3) = \{ \phi(e_2, e_3), \phi(d_2, d_3), \phi(e_2, e_4), \phi(e_2, e_5), \phi(d_1, d_2) \}$$

already set

assign e_3



$P(e_3 | e_u)$ with no cyclic rel.
 we have $[e_2, e_3] \subseteq [e_u]$ $e_2 < e_1$
 we can have $e_2 \rightarrow e_u \rightarrow e_3$



We go through vertices' level by level

- look for adjacent edge that goes to v
 check the vertices in L_{-1} that their eq class don't have an order.

↳ order them with no cyclic relation to all the vertices with edges part of v

- look for adjacent edge that goes from v .

(no cyclic order)

(work on a connected graph)



* for k

1) no incoming adj. edges to k

2) $(k, g), (k, f) \rightarrow L = \{g, f\}$

↳ loop of L

↳ g loop over what's before?

↳ $order = [g]$

$\hookrightarrow$ loop over what's before
 if $(eq(g, i)) = \emptyset$
 $\varphi(eq(g, i)) = 1$ arbitrary ($\varphi = 1$ for this case)
 $order = [g, i]$ $\varphi(eq(g, i)) = 1$

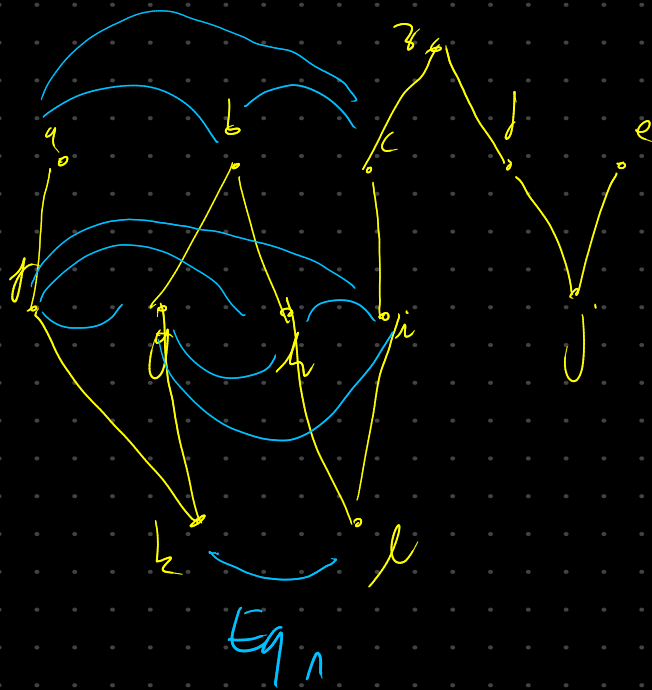
$- l^0 \Rightarrow \varphi(E_{g,i}) = 0$
 1) no incoming adj edges to l
 2) $(l, h), (l, i) \forall i \in eq(l, h) \quad i \in \{0, 1\}^2$
 $L = \{h, i\}$

$\Rightarrow$ loop over L
 $\hookrightarrow$ loop over what's before
 $order = [h]$
 $\hookrightarrow \varphi(eq(h, i)) = \varphi(E_{h,i}) = 0$

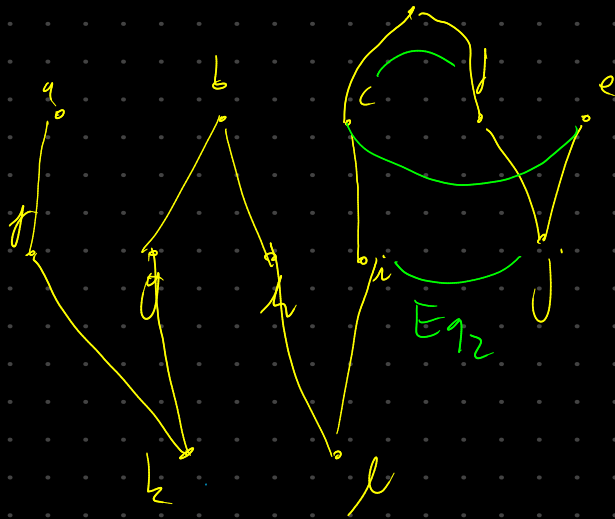
$\Rightarrow$ move to l_2

$- j^1 \times$

2) $L = \{d, e\} \Rightarrow order = [d, e] \quad \varphi(eq(d, e)) = 1$



$$E_{q_1} = \left\{ \begin{array}{l} a, d, (a,b), (b,c), \\ (f,g), (f,h), (g,h), (g,i), \\ (h,g), (h,i), \\ (k,l), \bar{l} \end{array} \right\}$$

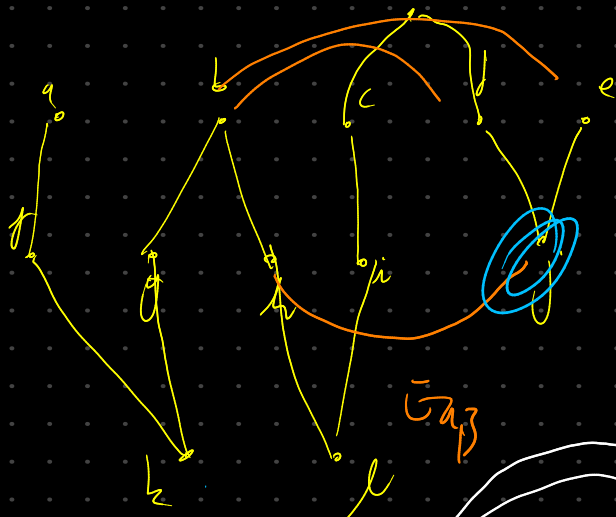


$$E_{q_2} = \{ (c,d), (h,i), \bar{y} \}$$

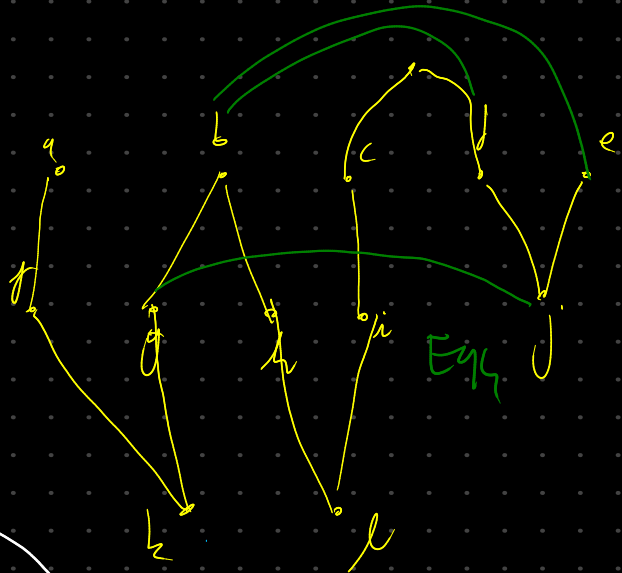
$$E_{q_3} = \{ (b,d), (b,e), (h,i) \}$$

$$E_{q_4} = \{ (g,i), (b,l), (b,e) \}$$

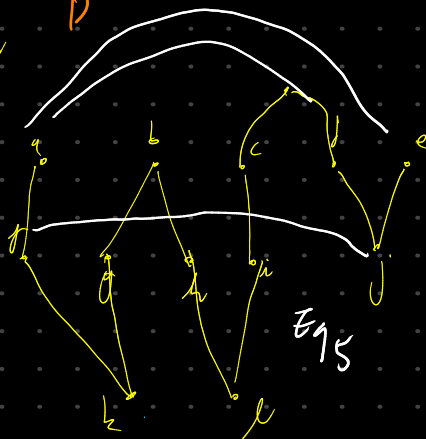
$$E_{q_5} = \{ (a,d), (a,e), (f,i) \}$$



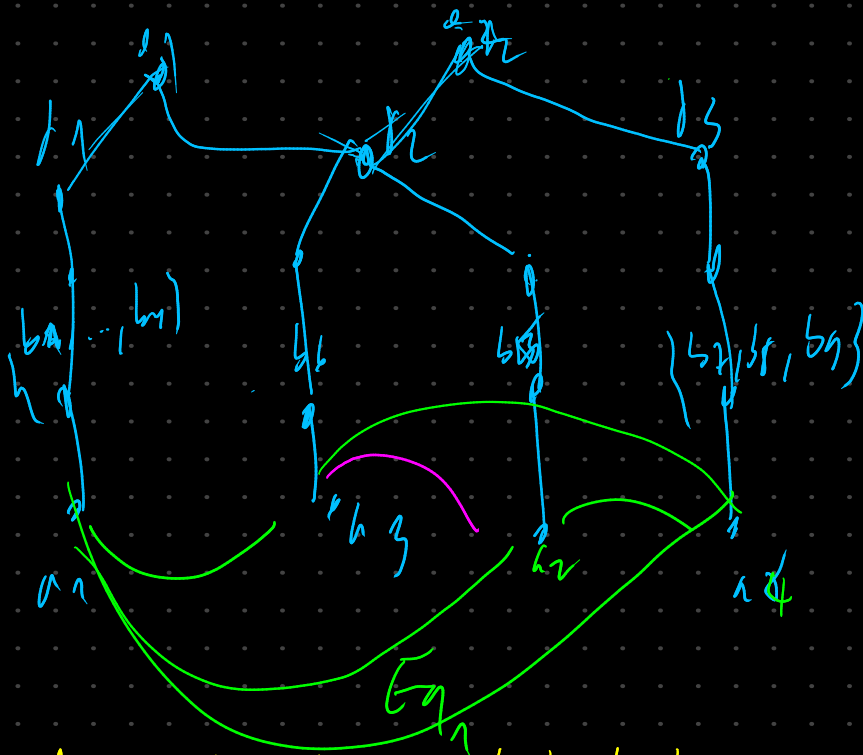
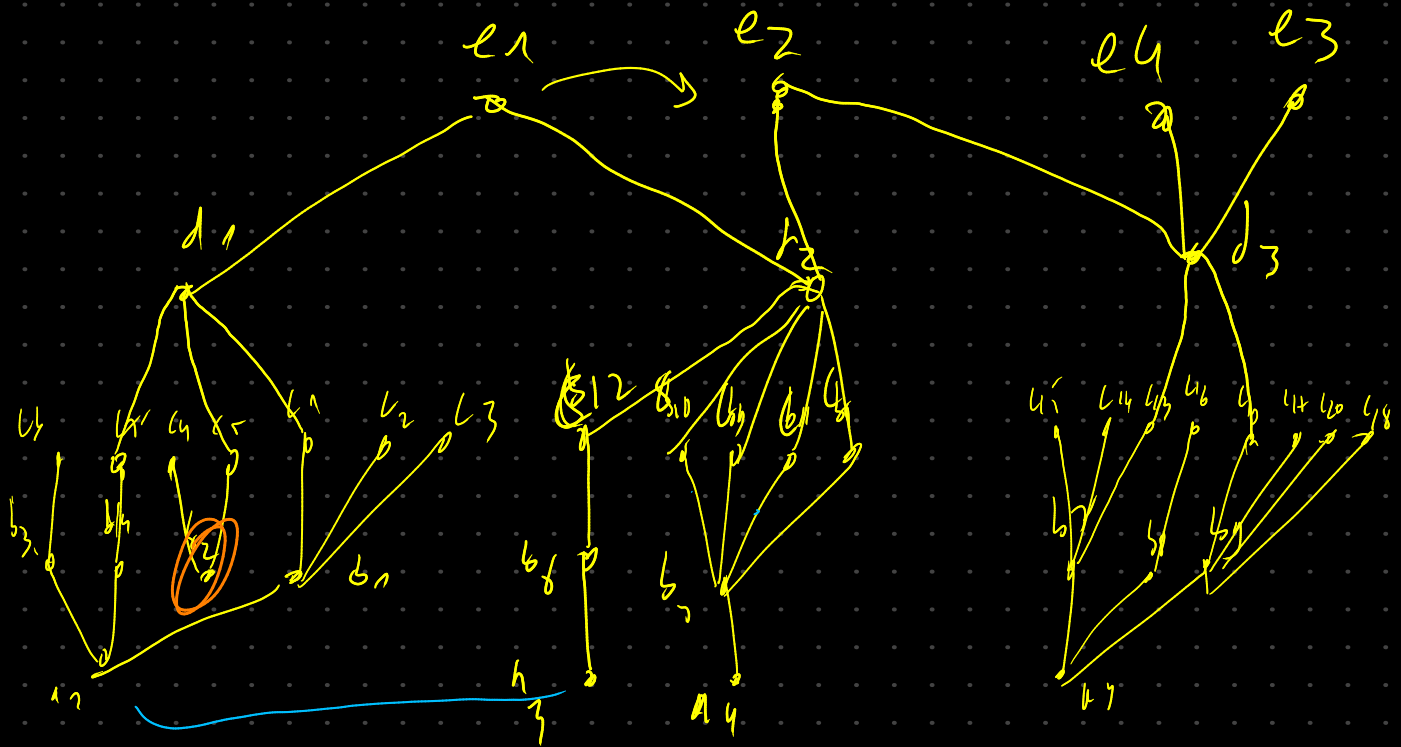
$\bar{E}_{q_3}$



E_{q_4}



E_{q_5}

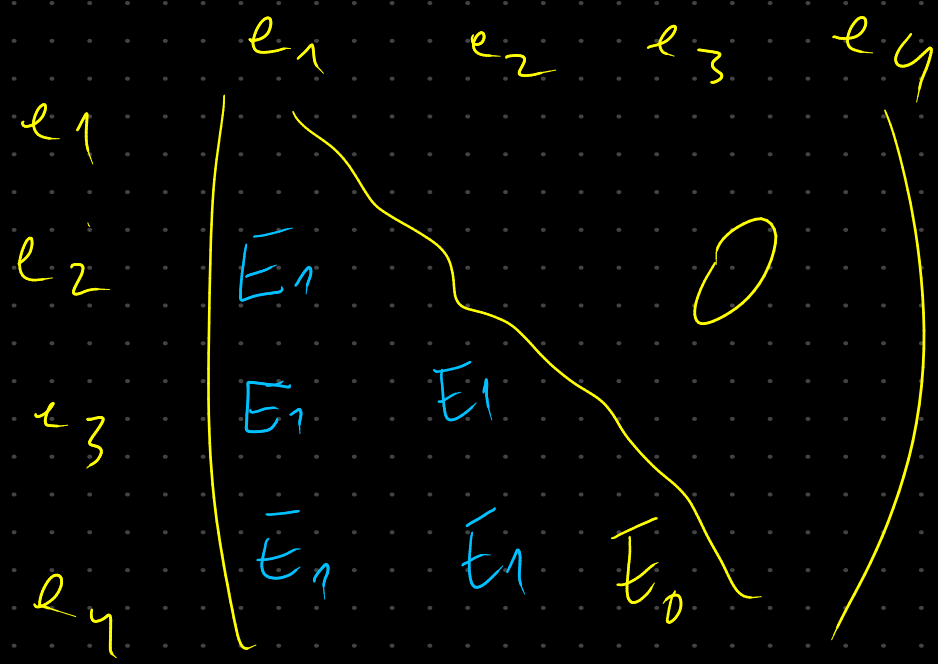
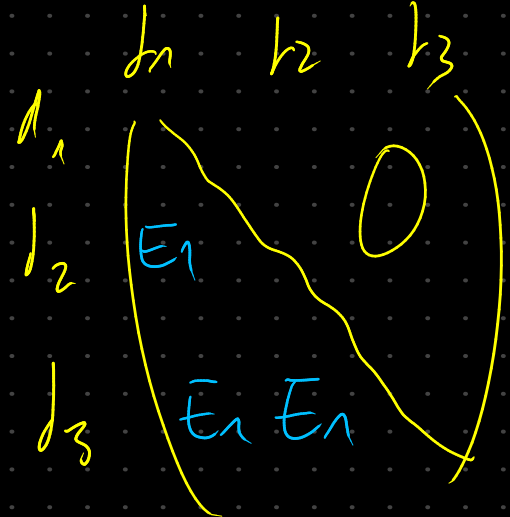
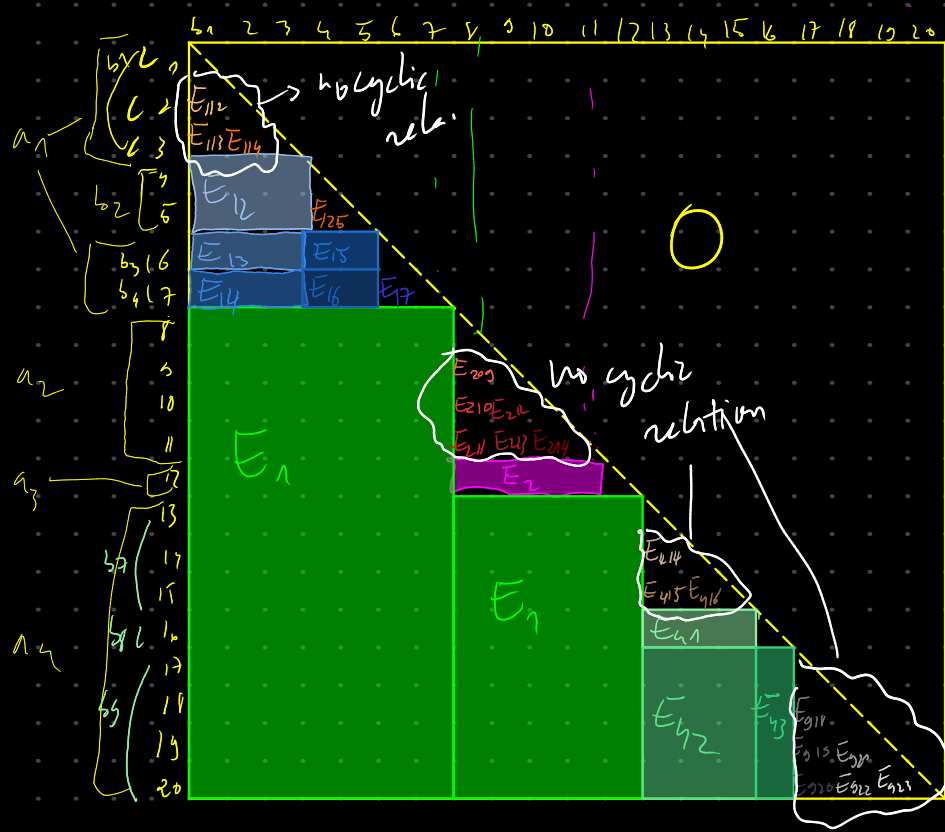


	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9
a_1	b_1								
a_2	b_2	E_{12}							
a_3	b_3	E_{13}	E_{15}						
a_4	b_4	E_{14}	E_{16}	E_{17}					
a_5	b_5	E_{18}	E_{19}	E_{20}	E_{21}				
a_6	b_6	E_{22}	E_{23}	E_{24}	E_{25}	E_{26}			
a_7	b_7	E_{27}	E_{28}	E_{29}	E_{30}	E_{31}	E_{32}		
a_8	b_8	E_{33}	E_{34}	E_{35}	E_{36}	E_{37}	E_{38}	E_{39}	
a_9	b_9	E_{40}	E_{41}	E_{42}	E_{43}	E_{44}	E_{45}	E_{46}	E_{47}

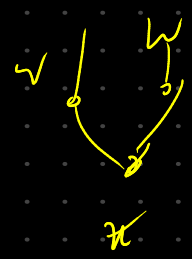
no cyclic relation

no cyclic relation

	a_1	a_2	a_3	a_4
a_1				
a_2	E_1			
a_3	E_1	E_2		
a_4	E_1	E_1	E_1	



- find vertices v with no incoming edges and
 $\exists w, w' \in V(h)$ with $l(w) = l(w') = l(v)$ and
 $\exists x \in V(h)$ with $(x, w), (x, w') \in E(h)$



$\exists v \mapsto$ problematic

Save φ classes from v to the vertices
of the same group of adjacent
edges.

choose arbitrary an eq. class and assign it
without adding a cyclic relation for vertices
with adjacent edges.

iteration 1: $\phi(E_1) = 1$

iteration 2: $\phi(E_{15}) = 1 \Rightarrow \phi(E_{16}) = \phi(E_{12}) = 1$

iter. 3: $\phi(E_{14}) = 1$

iter 4: $\phi(E_{17}) = 1$

we have $\phi(E_{14}) = 1 \begin{pmatrix} b_1 & b_4 \\ \cdot & \cdot \end{pmatrix}$
 $\phi(E_{17}) = 1 \begin{pmatrix} b_3 & b_7 \\ \cdot & \cdot \end{pmatrix} \Rightarrow$ We can pick
 whatever
 for $\phi(E_{13}) = 1$
 $\begin{pmatrix} b_1 & b_3 \\ \cdot & \cdot \end{pmatrix}$

iter 5: $\phi(E_2) = 0$

iter 6: $\phi(E_{918}) = 0$

we can fill all the eq. classes
that can lead to a cyclic relation
by making them equal each other

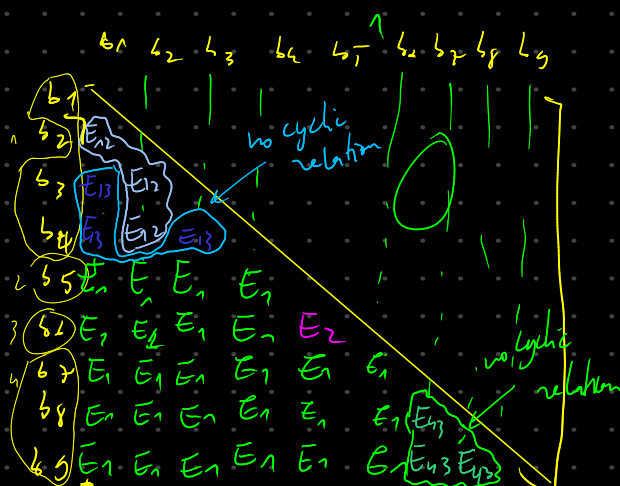
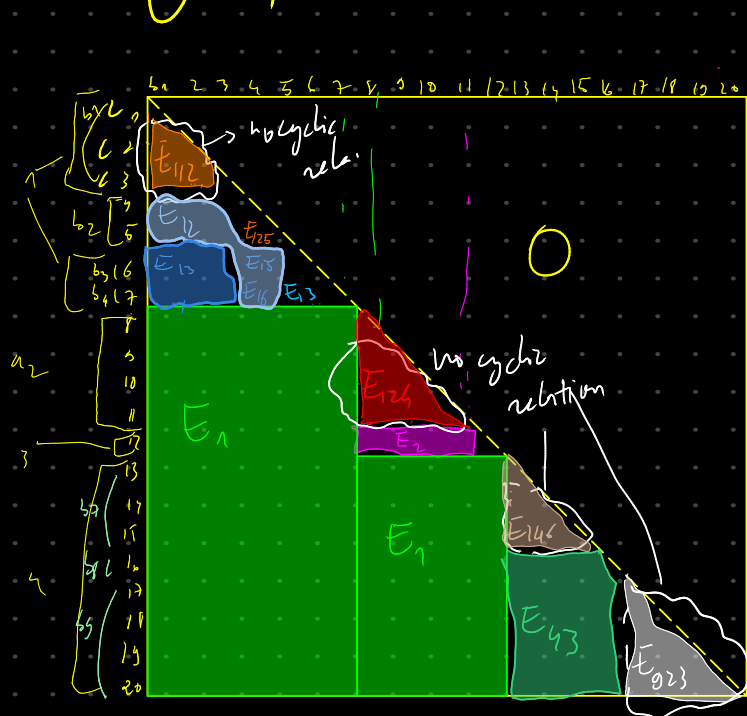


$$\begin{array}{l}
 E_{g18} \\
 E_{g15} \bar{E}_{g20} \\
 E_{g20} \bar{E}_{g22} \bar{E}_{g23}
 \end{array}
 \begin{array}{l}
 P(E_{g18}) = P(E_{g15}) = P(E_{g20}) = P(\bar{E}_{g22}) = P(E_{g22}) \\
 = P(E_{g23}) \\
 = 0
 \end{array}$$

iteration 7.

$$P\left(\begin{array}{c} E_{41} \\ E_{42} \bar{E}_{43} \end{array}\right) = 1$$

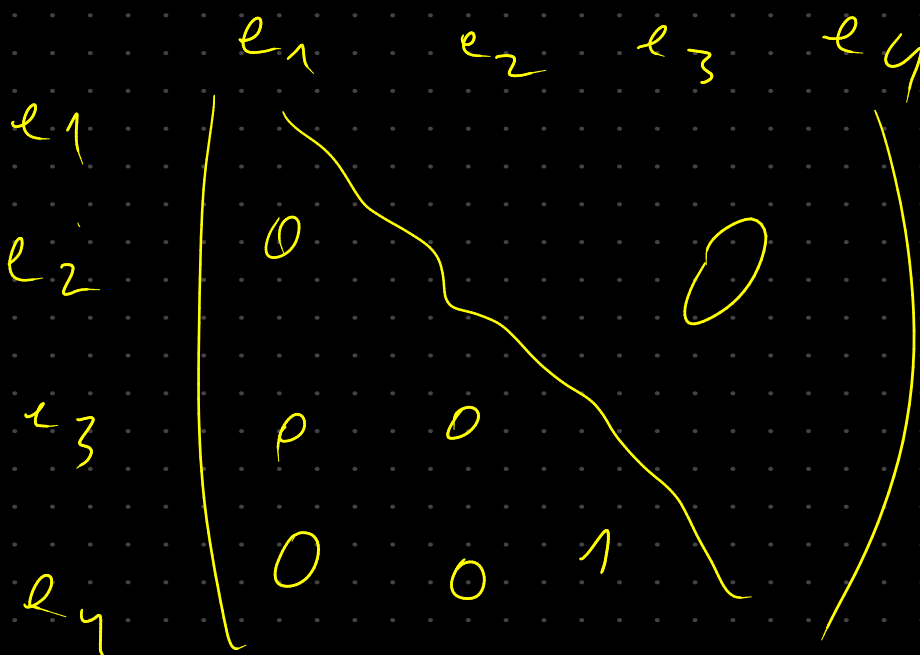
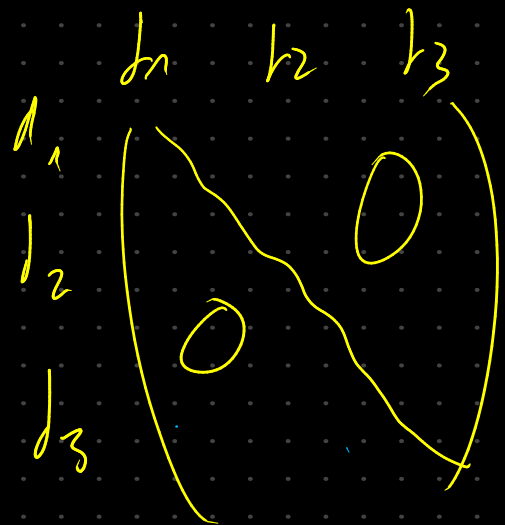
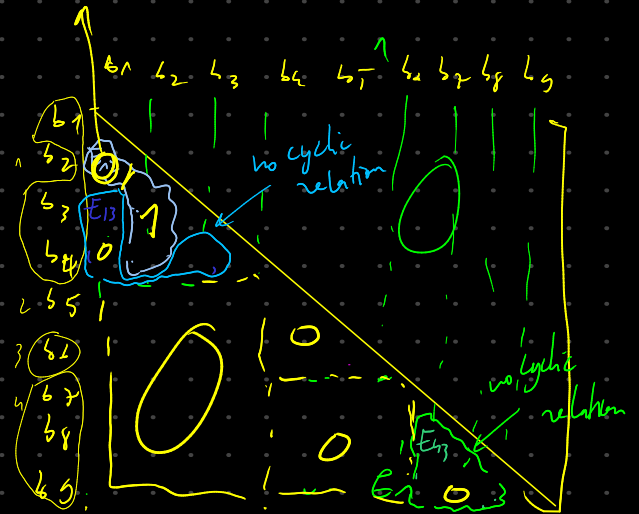
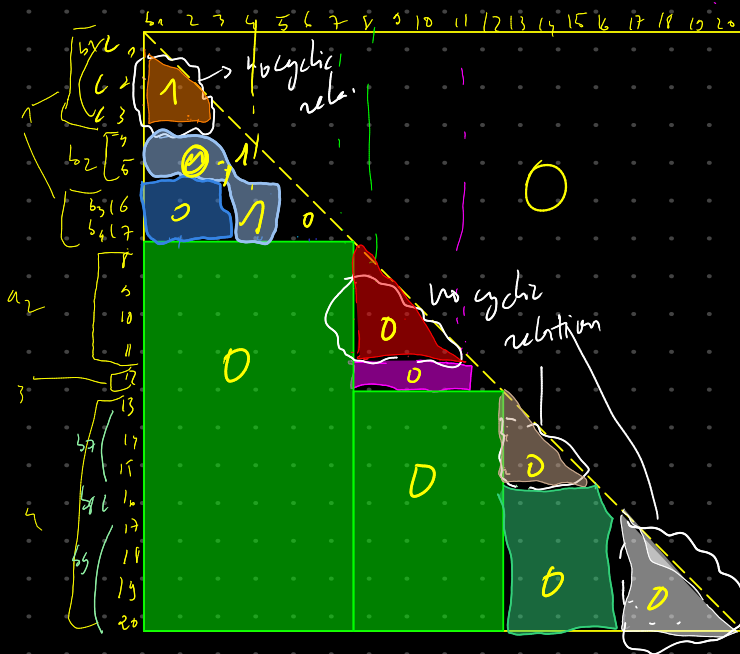
actually, we can reduce the equivalent classes

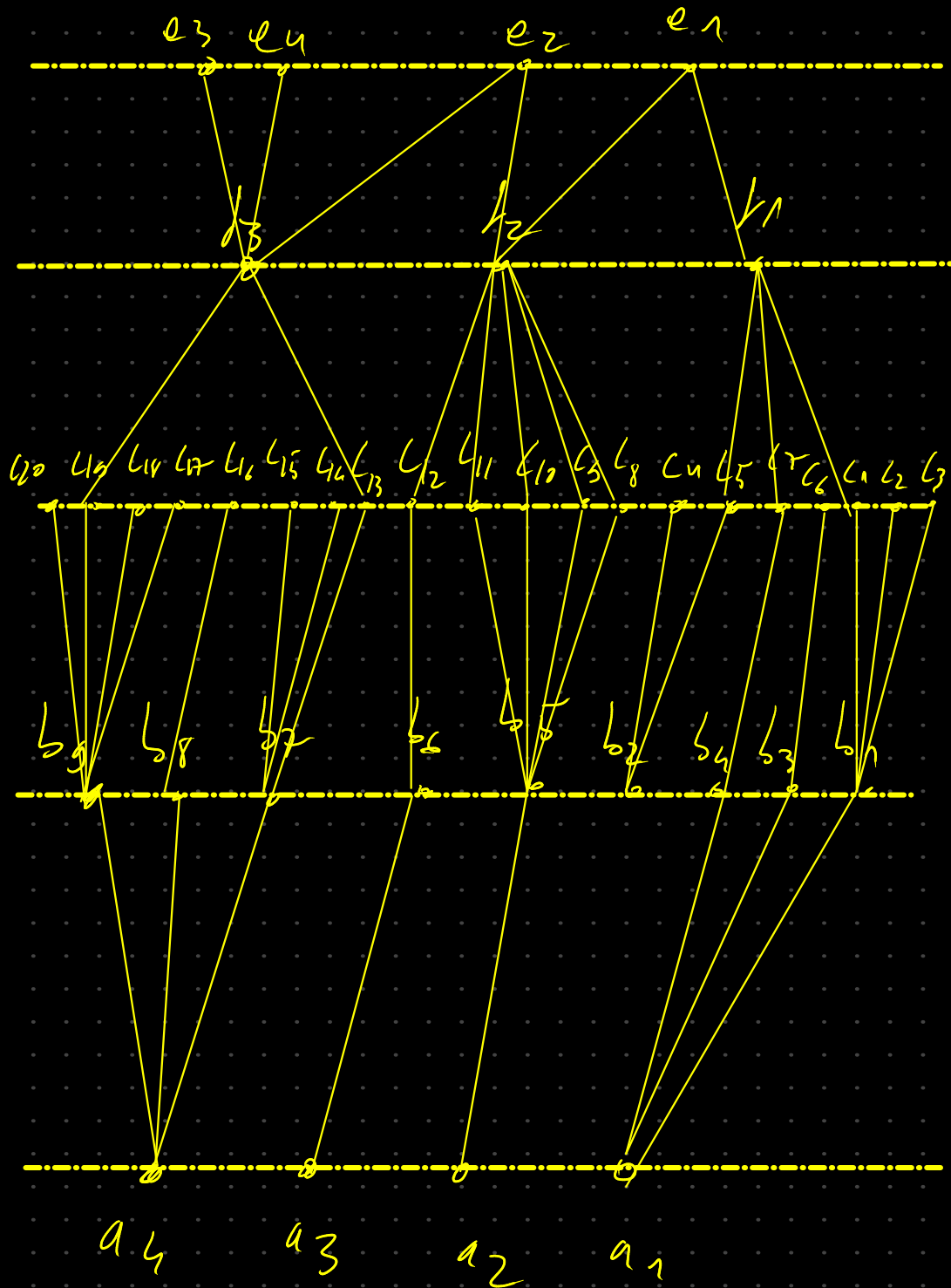


eq class: (we can give arbitrary assignment to them)

$$\begin{array}{cccccccccccc}
 E_1 & E_2 & E_{12} & E_{13} & E_{43} & E_{12} & E_{125} & E_{124} & E_{146} & E_{223} & E_0 \\
 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1
 \end{array}$$

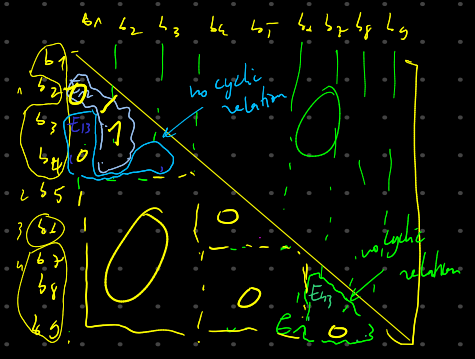
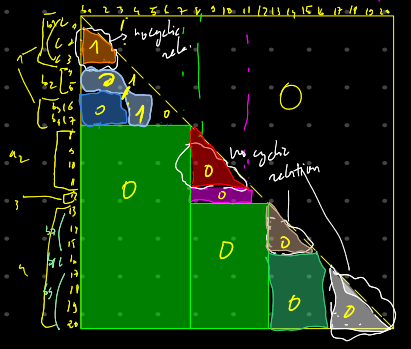
$$\varphi(l_2, l_1) = 1 = \varphi(l_2, b_3) = \varphi(l_2, l_4)$$



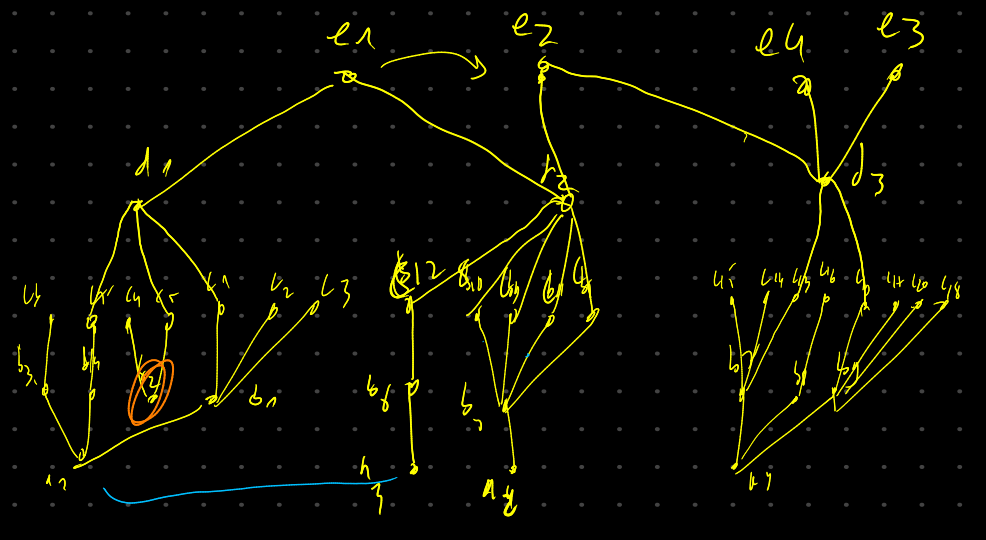


$$\begin{matrix} e_1 & e_2 & e_3 & e_4 \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

$$\begin{matrix} l_1 & l_2 & l_3 \\ \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{matrix}$$



$$\begin{matrix} a_1 & a_2 & a_3 & a_4 \\ \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \end{matrix}$$



Algo:

group the vertices that are the disjoint vertices of adjacent edges coming from level below them.

concatenate the equivalent classes of these vertices into one equivalent class for each group

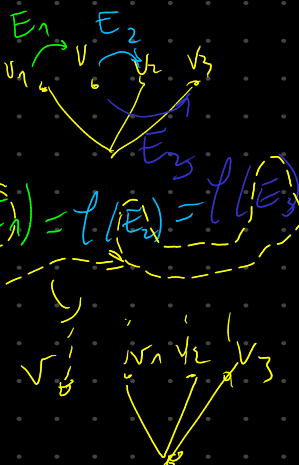
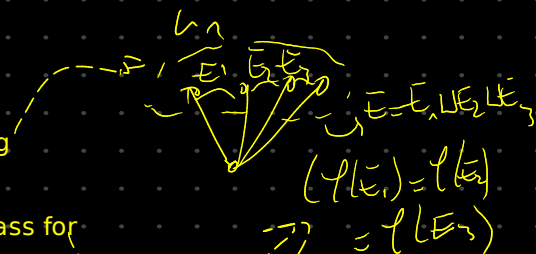
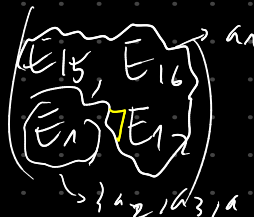
find vertices that don't have an incoming edge

for a vertex v that doesn't have an incoming edge:
check vertices from the same level that are part of one these groups

concatenate all the equivalent classes of these vertices
into

(before) v (after)

$$\mathcal{P}(\mathcal{T}E_v) = \mathcal{P}(E_A)$$



in the example:

$$\mathcal{T}E_{12} = E_{15} = E_{16} = E_1$$

fix the previous example in accordance to the Algo.

